



Legal Saathi (لیگل ساتھی)

AI-Powered Bilingual DIY Legal Document Generation Platform for Pakistan

Architectural specification, statutory knowledge base, REST API catalog, and Electronic Transactions Ordinance (ETO 2002) compliance report.

Project Title & Tagline	Legal Saathi — Democratizing Court-Compliant Legal Access in Urdu (Nastaliq) & English
Web Client (Frontend)	https://legal-saathi-one.vercel.app LIVE ON VERCEL EDGE
REST API (Backend)	http://legalsaathi.runasp.net LIVE ON MONSTERASP
Swagger Documentation	http://legalsaathi.runasp.net/swagger (OpenAPI 3.0 Interactive Explorer)
Cloud Database	Microsoft SQL Server 2022 (db67009.databaseasp.net) with Pure ADO.NET Stored Procedures
Automated QA Status	23 / 23 Tests Passing (100%) across Auth, Cryptography, Database & AI Fallback
Repository Security	Zero Exposed Secrets • Local .env Isolation with Custom DotEnvLoader Architecture

1 Problem Statement & 4-in-1 Solution Pillars

1. Bilingual DIY Wizard

Generates court-compliant contracts in Urdu (Nastaliq) & English via guided step-by-step questionnaires.

2. Statutory Legal AI

Dual-tier engine (Groq LLaMA 3.3 70B + offline rule engine) trained on core Pakistani statutes.

3. ETO 2002 Signatures

SHA-256 tamper-evident digital signatures with SMS/Email OTP verification valid in Pakistani courts.

4. Stamp Duty Engine

Calculates provincial stamp duties & e-Stamping values across Punjab, Sindh, KPK, ICT, & Balochistan.

2 End-to-End System Architecture

Decoupled Enterprise Multi-Tier Stack

Sub-millisecond data execution, client-side caching, and zero-ORM runtime overhead.



Presentation Layer

- Next.js 14 App Router (React 18)
- TypeScript 5.x & Tailwind CSS
- Bidirectional Nastaliq Typography
- Client-Side CNIC & Phone Masking



API Orchestration

- ASP.NET Core 8 Web API (C#)
- JWT Bearer + Secure HttpOnly Cookie
- Centralized Security Middleware
- Clean Modular Architecture



High-Perf Data Engine

- Cloud Microsoft SQL Server 2022
- 100% Parameterized ADO.NET SPs
- SHA-256 Cryptographic Audit Logs
- QuestPDF (RTL Vector) & OpenXML

3 Complete RESTful API Endpoint Specification

All endpoints are secured via JWT authorization policies, rate-limited, and documented in the OpenAPI 3.0 specification:

Method	Endpoint Route	Description & Workflow	Auth Policy
POST	/api/auth/register	Register citizen, corporate admin, or advocate with OTP dispatch	Anonymous
POST	/api/auth/verify-email	Verify 6-digit OTP and issue JWT access token & secure cookies	Anonymous
POST	/api/auth/login	Authenticate user with email/phone & PBKDF2 hashed password	Anonymous
POST	/api/auth/refresh-token	Exchange refresh token for a fresh JWT session seamlessly	Anonymous
GET	/api/templates	List court-compliant legal templates by category & tier	Public
GET	/api/templates/{slug}	Retrieve dynamic questionnaire schema & bilingual contract clauses	Public
POST	/api/documents/generate	Compile form answers, calculate stamp duty, and produce PDF/DOCX	RequireEndUser
GET	/api/documents/user	Fetch user's generated agreements, draft status & download paths	RequireEndUser
POST	/api/signatures/sign	Execute ETO 2002 compliant signature with SHA-256 hash stamp	RequireEndUser
POST	/api/ai/ask	Inquire with Pakistani Legal AI with document context & Urdu NLP	Public / User
GET	/api/admin/stats	Aggregate platform statistics (users, docs, lawyers, stamp duty)	RequireSuperAdmin
PUT	/api/admin/lawyers/{id}/verify	Verify advocate Bar Council credentials and grant certified status	RequireSuperAdmin

4 Database Schema & Relational Entity Structure

Normalized 3NF relational schema enforcing strict referential integrity, encrypted credentials, and temporal timestamps:

dbo.Users & dbo.Roles

PK: UserID (int)
Fields: Name, Email, Phone, PasswordHash (PBKDF2), Role_ID (FK), Cnic, IsEmailVerified, CreatedDateTime.
Roles: 1: SuperAdmin, 2: EndUser, 3: CorporateAdmin, 4: Lawyer.

dbo.Templates & FormFields

PK: TemplateID (int)
Fields: Slug, TitleEng, TitleUrdu, ContentTemplateEng, ContentTemplateUrdu, BasePrice, EstimatedStampDuty.
Fields Table: Dynamic questionnaire types (Text, CNIC, Phone, Currency, Date).

dbo.UserDocuments

PK: UserDocumentID (int)
Fields: User_ID (FK), Template_ID (FK), FormAnswersJson, StatusID, DocumentHash (SHA-256), StoragePath, DocxPath.
Integrity: Stores full questionnaire payload.

dbo.Signatures

PK: SignatureID (bigint)
Fields: UserDocument_ID (FK), SignerName, SignerCnic, SignerPhone, Sha256Hash, IpAddress, UserAgent, Timestamp.
Compliance: ETO 2002 court evidence.

dbo.AIQueries

PK: QueryID (bigint)
Fields: User_ID, Prompt, Response, LanguageCode, Tokens, ModelUsed, CreatedDateTime.
Analytics: Tracks token usage & user intent across Pakistani legal topics.

dbo.AuditLogs

PK: AuditLogID (bigint)
Fields: User_ID, Action, EntityName, EntityID, IPAddress, UserAgent, IntegrityChecksum, CreatedDateTime.
Security: Immutable append-only audit trail.

5 Pakistani Statutory AI Knowledge Codex

📖 Core Statutory Frameworks Integrated

- **Contract Act 1872 & Specific Relief Act 1877:** Breach remedies, consideration, liquidated damages, indemnity clauses.
- **Transfer of Property Act 1882:** Conveyance deeds, commercial leases, mortgages, oral/written Gift Deeds (*Hiba*).
- **Succession Certificates Act 2021:** NADRA digital succession certificates for legal heirs without civil litigation.
- **Muslim Family Laws Ordinance 1961:** Nikahnama registration, Talaq, Khula, Dower/Mahr disputes, maintenance.
- **Provincial Tenancy Ordinances:** Punjab Rented Premises Act 2009 & Sindh Rented Premises Ordinance 1979 eviction procedures.

⚡ Dual-Tier Inference & Resilience Architecture

- **Primary Engine:** Groq Cloud LLaMA 3.3 70B delivering sub-second bilingual natural language inference.
- **Urdu NLP Support:** Native comprehension of Urdu Nastaliq text and Roman Urdu queries.
- **Offline Fallback Codex:** Built-in deterministic Pakistani legal rule engine ensures 100% platform availability even during external API outages or rate limits.
- **Legal Disclaimers & Ethics:** Recommends Bar-certified advocate review for contested litigation or high-value contracts.

6 Provincial Stamp Duty & e-Stamping Rates

Province / Territory	Governing Statutory Schedule	Calculated Duty Rates & Stamp Paper Requirements
Punjab	Board of Revenue Punjab (e-Stamping)	Affidavits: Rs. 100; Rent Deeds: 0.25% – 0.5% annual rent; NDAs: Rs. 100.
Sindh	Schedule I of Sindh Stamp Act 1899	Affidavits: Rs. 100; Tenancy: 2% of annual rent; Vehicle Sale: Rs. 500.
Khyber Pakhtunkhwa (KPK)	KPK Stamp (Amendment) Schedule	Affidavits: Rs. 100; Commercial Service Contracts: Ad-valorem sliding scale.
Islamabad (ICT)	ICT Stamp Duty Schedule	Affidavits: Rs. 100; Rent agreements: Rs. 500 fixed / ad-valorem.
Balochistan	Balochistan Board of Revenue	Non-judicial stamp paper with standard court revenue stamps.

7 Quality Assurance, Security & Verification Report

23 / 23

XUNIT TESTS PASSING
100% Success Rate in CI/CD

0 Keys

TRACKED IN SOURCE CONTROL
Strict .env Isolation

ETO 2002

COURT COMPLIANT
SHA-256 Tamper Evident

🚀 Verified Deployment Links & Resources

- **Live Web App:** <https://legal-saathi-one.vercel.app> (Vercel Serverless)
- **Backend API:** <http://legalsaathi.runasp.net> (MonsterASP Cloud)
- **Interactive Swagger Docs:** <http://legalsaathi.runasp.net/swagger>
- **GitHub Repository:** <https://github.com/CodeByArsalan/LegalSaathi>